

WASP-163b

R-2285

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_250610 · created 2026-08-17T00:25:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460836.827 +/- 0.05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.043
Transit depth (Rp/Rs)^2	2.07 +/- 1.23 [%]
Orbital Inclination (inc)	85.82 +/- 2.69
Airmass coefficient 1 (a1)	1.09 +/- 0.024
Airmass coefficient 2 (a2)	-0.061 +/- 0.064
Scatter in the residuals of the lightcurve fit is	2.22 %
Best Comparison Star	#3 - [201, 275]
Optimal Method	PSF photometry
Transit Duration (day)	0.074 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.144 ± 0.043 vs 0.119 (deviation 0.58σ)
Duration fit vs published	0.074 d vs 0.0942 d (deviation 0.75σ)
Mid-time O–C	-1.1 min
Depth SNR	3.3
Residual scatter	2.22 %

Light curve

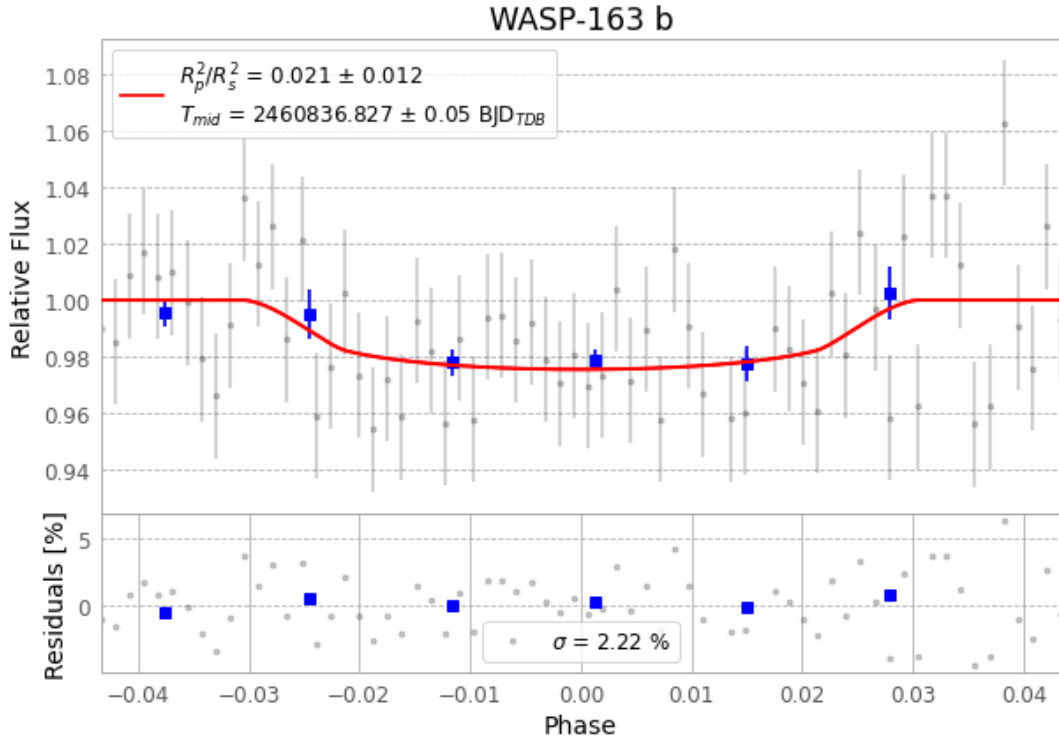


Figure 1 — FinalLightCurve_WASP-163 b_2025-06-09.png (SHA-256 b376e61388a93ef9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcee5f92b190a8ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

68 FITS frames (45 MB), WASP-163250610060015.FITS → WASP-163250610092115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-250610.log (b62fdf7fa2ae...), FinalLightCurve_WASP-163 b_2025-06-09.png (b376e61388a9...), FinalLightCurve_WASP-163 b_2025-06-09.csv (da7491f9b6d9...)

Compute resources

Wall time 102.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:25Z.